Leverage Index Project Documentation

Overview

This project calculates a Leverage Index (LI) for game situations and links it to pitcher performance metrics. The purpose is to compare pitcher KPIs across low-, medium-, and high-leverage plate appearances. Leverage is based on the expected change in win expectancy given the inning, half-inning, score differential, outs, and base-runner state. The Leverage Index model uses Power 5 game data from 2023–2025 to establish win expectancy and event probabilities. The KPI analysis applies the resulting LI values to the 2025 pitch and plate-appearance data.

Process

Part 1: Leverage Index Model

1. Load and prepare game data:
   Trackman play and pitch data, play-by-play outcomes, game/session information, and baserunner data are loaded from the database. The Leverage Index model uses plate-appearance-ending events, so each plate appearance has one final outcome.
2. Calculate the score before each play:
   Plays are sorted chronologically within each game. Runs scored are accumulated separately for the home and away teams, then shifted so the model uses the score before the current play. The home score minus the away score creates the score differential.
3. Determine the winner of each game:
   Final home and away scores are calculated for each game. A home-team-win indicator is added and serves as the outcome used to estimate win expectancy.
4. Create base-runner states:
   The baserunner columns are converted into eight possible base states: bases empty, each individual runner, each two-runner combination, and bases loaded.
5. Build the win expectancy table:
   Win expectancy is calculated for each game state using inning, half-inning, score differential, outs, and base state. The score differential is capped between -7 and +7 runs because larger leads have limited sample sizes and do not meaningfully affect the leverage calculation. Extra innings are grouped into an inning-10 bucket. The table uses smoothing, so less-common game states borrow information from broader score, inning, and base/out contexts instead of relying only on a small number of observations.
6. Calculate event probabilities:
   Plate-appearance outcomes are categorized as strikeout, other out, walk, single, double, triple, or home run. Non-event pitches are excluded. The frequency of each event is used as its probability in the leverage calculation.
7. Calculate Leverage Index:
   For each unique game state, the model simulates each possible event outcome. It determines the resulting game state, looks up the new win expectancy, and calculates the absolute change from the current win expectancy. These changes are weighted by the event probabilities to produce an expected

win-expectancy swing. Leverage Index equals the expected win-expectancy swing for a game state divided by the average expected swing across all states. An LI of 1.00 represents an average-leverage situation. Values above 1.00 represent above-average leverage, while values below 1.00 represent below-average leverage.

8. Validate and visualize the LI output:
   The highest-leverage states are reviewed along with their win-expectancy sample sizes. Additional visual checks show the LI distribution, average LI by inning and score differential, and average LI by base state and outs.

Part 2: Pitcher KPI Analysis

1. Load 2025 pitch-level data and connect LI values:
   The KPI analysis loads 2025 Trackman pitch and play-by-play data. The final LI value for each plate appearance is joined to all pitches within that plate appearance using game ID, inning, half-inning, and plate-appearance number.
2. Create configurable LI buckets:
   Each plate appearance is assigned to a Low, Medium, or High LI bucket based on LI percentiles. The current configuration uses equal thirds: bottom one-third, middle one-third, and top one-third of observations. The bucket definitions are configurable in a single location, so the percentages can be changed without revising downstream KPI calculations.
3. Create pitch-level KPIs:
   Pitch-level data is used to calculate Strike%, Heart%, Whiff%, CSW%, and Chase% for each pitcher within each LI bucket.
4. Create plate-appearance-level KPIs:
   The final event of each plate appearance is retained for plate-appearance metrics. K%, BB%, and Ideal Contact Rate are calculated by pitcher and LI bucket. Runner-only events, such as stolen bases and caught stealing, are excluded from plate-appearance calculations.
5. Calculate Run Value and Run Value per 100 pitches:
   Each pitch is matched to the run-value lookup using its pitch result, count, and outs. Batter run value is converted to pitcher run value by reversing the sign, so positive values are favorable for the pitcher. Total Run Value and Run Value per 100 pitches are calculated within each LI bucket.
6. Calculate wOBA allowed:
   wOBA is calculated at the plate-appearance level using the supplied weights. The weighted offensive value is divided by plate appearances to calculate wOBA allowed for each pitcher and LI bucket.
7. Combine KPI results and apply qualification thresholds:
   Pitch-level, plate-appearance-level, run-value, and wOBA results are combined into one pitcher KPI table. Minimum thresholds for pitches, plate appearances, and balls in play are applied before the data is used in charts or leaderboards.
8. Create visualizations:
   The analysis includes a selected pitcher profile across Low, Medium, and High LI buckets with team reference values; a High-LI pitcher bubble chart comparing wOBA allowed, RV per 100 pitches, K%, and sample size; a Low-to-High LI wOBA comparison chart; and a High-LI KPI heatmap comparing each pitcher's performance relative to the team.